EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

TERMINOLOGY OF COORDINATION COMPOUND

1. In the complex ion $[Fe(EDTA)]^-$ the coordination number and oxidation state of central metal ion is :-
(1) C.N. = 6 O.N. = +3
(2) C.N. =1 O.N. = –1
(3) C.N. = 4 O.N. = +2
(4) C.N. = 3 O.N. = +3

CC0001

2. The coordination number and oxidation number of the central metal ion in the complex $[Pt(en)_2]^{+2}$ is :-
(1) C.N. = 2, O.N. = +2
(2) C.N. = 6, O.N. = +4
(3) C.N. = 4, O.N. = +4
(4) C.N. = 4, O.N. = +2

CC0002

3. Select bidentate or didentate ligand from the following .
(1) CO (2) SCN^-
(3) CH_3COO^- (4) $C_2O_4^{2-}$

CC0003

4. The oxidation and coordination number of Pt in $[Pt(C_2H_4)Cl_3]^-$ is respectively :-
(1) + 1, 3 (2) + 2, 4
(3) + 3, 6 (4) + 2, 5

CC0004

5. The CN and ON of X in the compound $[X(SO_4)(NH_3)_5]$ will be :-
(1) 10 and +3 (2) 1 and +6
(3) 6 and +4 (4) 6 and +2

CC0005

6. What is the oxidation number of Fe in $[Fe(H_2O)_5(NO]^{2+}$ ion?
(1) +2 (2) +3
(3) +1 (4) 0

CC0006

7. Which one is a heteroleptic complex ?
(1) Ferrocene
(2) Chromocene
(3) Prussian blue
(4) Zeise's salt

CC0231

8. Select the set representing correct statements only.
(I) A coordination entity constitutes a central metal atom or ion bonded to a fixed number of ions or molecules.
(II) Ligand having two different or similar donor atoms in the complex is called bidentate ligand.
(III) All bidentate ligands are chelating ligands.
(IV) Ni^{+2}, Co^{+3} and Fe^{+3} are acting as Lewis acids in $[NiCl_2(H_2O)_4]$, $[CoCl(NH_3)_5]^{+2}$ and $[Fe(CN)_6]^{-3}$.
(1) I, II, III and IV (2) II and III
(3) I, III and IV (4) I, II and IV

CC0232

9. Oxidation number of Cobalt in $K[Co(CO)_4]$ is :-
(1) +1 (2) –1 (3) 0 (4) +2

CC0233

10. Which of the following species is not expected to be ligand :-
(1) NO (2) NH_4^+
(3) $NH_2CH_2CH_2NH_2$ (4) CO

CC0234

11. Identify the statement which is not correct :-
(1) Coordination compounds are mainly known for transition metals.
(2) Coordination number and oxidation state of central metal atom in a complex is same.
(3) Ligand donates at least one electron pair to CMA
(4) $[Co(NH_3)_4Cl_2]^+$ is a heteroleptic complex.

CC0235

12. Which of the following ligands will not show chelation?
(1) EDTA (2) DMG
(3) Ethane-1, 2 diamine (4) SCN^-

CC0236

IUPAC - NOMENCLATURE

13. $K_3[Fe(CN)_6]$ is :-
(a) Potassium hexacyano ferrate (II)
(b) Potassium hexacyano ferrate (III)
(c) Potassium ferri-cyanide
(d) Hexacyano ferrate (III) potassium
Correct answer is :-
(1) Only (a) and (b) (2) Only (b) and (c)
(3) Only (a) and (c) (4) Only (b) and (d)

CC0009

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14. Give the IUPAC name of the complex compound $[Co(NH_3)_4(H_2O)Br](NO_3)_2$.
(1) Bromoaquotetraamine Cobalt (III) nitrate
(2) Bromoaquotetraaminocobalt (III) nitrate
(3) Bromoaquatetraammine cobalet (III) nitrate
(4) Tetraammineaquabromido cobalt (III) nitrate
CC0010

15. Which of the following complex is anion :-
(1) Fluoro pentaammine cobalt(III)
(2) Trioxalato ferrate(III)
(3) Penta Carbonyl iron(0)
(4) Dichloro diammine platinum(II)
CC0011

16. The chloro-bis (ethylenediamine) nitro cobalt (III) ion is :-
(1) $[Co(NO_2)_2(en)_2Cl_2]^+$ (2) $[CoCl(NO_2)_2(en)_2]^+$
(3) $[CoCl(en)_2NO_2]^+$ (4) $[Co(en)Cl_2(NO_2)_2]^-$
CC0012

17. Which of the following complexes is not a chelate
(1) bis (dimethylglyoximato) nickel(II)
(2) Potassium ethylenediaminetetrathiocyanato chromate(III)
(3) Tetrammine dichlorocobalt(III) nitrate
(4) Trans–diglycinatoplatinum(II)
CC0013

18. The correct IUPAC name of the complex $[Fe(\eta^5\text{–}C_5H_5)_2]$ is
(1) Cyclopentadienyl iron (II)
(2) Bis (cyclopentadienyl) iron (II)
(3) Dicyclopentadienyl ferrate (II)
(4) Ferrocene
CC0014

19. The correct name of $[Pt(NH_3)_4Cl_2]$ $[PtCl_4]$ is :-
(1) Tetraammine dichloro platinum (IV) tetrachloro platinate (II)
(2) Dichloro tetra ammine platinium (IV) tetrachloro platinate (II)
(3) Tetrachloro platinum (II) tetraammine platinate(IV)
(4) Tetrachloro platinum (II) dichloro tetraamine platinate (IV)
CC0015

20. The IUPAC name of $K_2[Cr(CN)_2O_2(O)_2(NH_3)]$ is:-
(1) Potassiumamminedicyano dioxoperoxochromate(VI)
(2) Potassiumamminecyanoperoxodioxo chromium (VI)
(3) Potassiumamminecyanoperoxodioxo chromium (VI)
(4) Potassiumamminecyanoperoxodioxo chromate (IV)
CC0016

21. The IUPAC name for $[Co(NH_3)_6]$ $[Cr(CN)_6]$ is :-
(1) Hexaammine cobalt (III) hexacyanochromate (III)
(2) Hexacyanochromium cobalt hexaammine (VI)
(3) Hexaammine cobalt (III) hexacyanochromium (VI)
(4) Hexacyanochromium (III) hexaammine cobalt (III)
CC0017

22. The IUPAC name for $[Co(NCS)(NH_3)_5]Cl_2$ is :-
(1) Pentaammine (thiocyanato-N) cobalt (III) chloride
(2) Pentaammine (thiocyanato-S) cobalt (III) chloride
(3) Pentaammine (isothiocyanato-N,S)cobalt (III) chloride
(4) Pentaammine (mercapto-N) cobalt (III) chloride
CC0018

23. Which rule is incorrect regarding IUPAC nomenclature of complex ions ?
(1) Cation is named first and than anion is named
(2) In the coordination sphere, the ligands are named alphabetically.
(3) Positively charged ligands have suffix-ate.
(4) More than one same type of ligands are indicated by di, tri, tetra etc.
CC0237

WERNER'S THEORY & EAN

24. Which of the following has least conductivity in aqueous solution.
(1) $CoCl_3 \cdot 4NH_3$ (2) $CoCl_3 \cdot 3NH_3$
(3) $CoCl_3 \cdot 5NH_3$ (4) $CoCl_3 \cdot 6NH_3$
CC0020

25. If EAN of a central metal ion X^{+2} in a complex is 34. and atomic number of X is 28. The number of monodentet ligands present in complex are:-
(1) 3 (2) 4 (3) 6 (4) 2
CC0021

26. The EAN of cobalt in the complex ion $[Co(en)_2Cl_2]^+$ is :-
(1) 27 (2) 36 (3) 33 (4) 35
CC0022

27. The effective atomic number of Cr (atomic no. 24) in $[Cr(NH_3)_6]$ Cl_3 is
(1) 35 (2) 27 (3) 33 (4) 36
CC0023

28. Which gives only 25% mole of AgCl, when reacts with $AgNO_3$:-
(1) $PtCl_2.4NH_3$ (2) $PtCl_4.5NH_3$
(3) $PtCl_4.4NH_3$ (4) $PtCl_4.3NH_3$
CC0024

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29. In the metal carbonyls of general formula $M(CO)_x$ (Which follows EAN rule) if M is Ni, Fe and Cr the value of x will be respectively:-
(1) 6, 5, 6 (2) 4, 5, 6 (3) 4, 4, 5 (4) 4, 6, 6
CC0025

30. A compound has the empirical formula $CoCl_3.5NH_3$. When an aqueous solution of this compound is mixed with excess silver nitrate, 2 mol of AgCl precipitate per mol of compound. On reaction with excess HCl, no NH_4^+ is detected. Hence it is
(1) $Co(NH_3)_5Cl_2]Cl$ (2) $[Co(NH_3)_5Cl]Cl_2$
(3) $[Co(NH_3)_5Cl_3]$ (4) $[Co(NH_3)_4Cl_2]Cl.NH_3$
CC0026

31. Which is incorrect regarding Werner theory :-
(1) Co-ordination compounds metals show two types of valancy. @neetquestionpaper
(2) Primary valency is satisfied by negative ions.
(3) Secondary valency is satisfied by only neutral molecules.
(4) Secondary valency is responsible for geometry of complex.
CC0238

32. Number of electrolytes in the complex $CoCl_3.5NH_3$
(1) 3 (2) 2 (3) 4 (4) 5
CC0239

VALENCE BOND THEORY

33. Which of the following compound is paramagnetic
(1) Tetracyanonickelate (II) ion
(2) Tetraamminezinc (II) ion
(3) Hexaamine chromium (III) ion
(4) Diammine silver (I) ion
CC0028

34. The shape of the complex $[Ag(NH_3)_2]^+$ is :
(1) Octahedral (2) Square planar
(3) Tetrahedral (4) Linear
CC0029

35. Hexafluoroferrate (III) ion is an outer orbital complex. The number of unpaired electrons are
(1) 1 (2) 5 (3) 4 (4) 3
CC0030

36. The shape of $[Cu(NH_3)_4]SO_4$ is :
(1) Square planar (2) Pyramidal
(3) Octahedral (4) Tetrahedral
CC0031

37. Among the following ions, which one has the highest paramagnetism ?
(1) $[FeF_6]^{3-}$ (2) $[Fe(H_2O)_6]^{2+}$
(3) $[Cu(H_2O)_6]^{2+}$ (4) $[Zn(H_2O)_6]^{2+}$
CC0032

38. In the complex ion ML_6^{n+}, M^{n+} has five d-electrons and L is weak ligand then the magnetic properties of the complex ion correspond to how many unpaired electrons
(1) 0 (2) 5 (3) 2 (4) 3
CC0033

39. A magnetic moment of 1.73 BM will be shown by one among of the following compounds .
(1) $[Cu(NH_3)_4]^{2+}$ (2) $[Ni(CN)_4]^{2-}$
(3) $TiCl_4$ (4) $[CoCl_6]^{3-}$
CC0034

40. The magnetic property and the shape of $[Cr(NH_3)_6]^{3+}$ complex ions are :
(1) Paramagnetic, Octahedral
(2) Diamagnetic, square planar
(3) Paramagnetic, tetrahedral
(4) None of the above
CC0035

41. Amongest the following ions which one is high spin complex.
(1) $[Cr(NH_3)_6]^{2+}$ (2) $[Fe(H_2O)_6]^{3+}$
(3) $[Cu(H_2O)_6]^{2+}$ (4) $[Zn(H_2O)_6]^{2+}$
CC0036

42. Which of the following complexes is an inner orbital complex ?
(1) $[CoF_6]^{3-}$ (2) $[FeF_6]^{3-}$
(3) $[Cr(NH_3)_6]^{3+}$ (4) $[Fe(H_2O)_6]^{2+}$
CC0037

43. What is Incorrect for $K_4[Fe(CN)_6]$
(1) O.N of Iron is +2
(2) It exhibit diamagnetic character
(3) It exhibit paramagnetic character
(4) It involves d^2sp^3 hybridisation
CC0039

44. In which of the following molecules, central atom used orbitals of different principle quantum number in the hybridisation :-
(1) $[Fe(CO)_5]$ (2) $[Zn(NH_3)_6]^{+2}$
(3) $[Ni(CO)_4]$ (4) $[Ni(NH_3)_6]^{+2}$
CC0040

45. What are the geometric shape and the oxidation number of the copper atom, respectively, for the complex ion, $[Cu(NH_3)_4(OH_2)_2]^{2+}$?
(1) Tetrahedral; + 2
(2) Square planar; – 2
(3) Linear; + 3
(4) Octahedral; + 2
CC0041

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46. For $[FeF_6]^{3-}$ and $[Fe(CN)_6]^{3-}$ magnetic moment of the fluoride complex is expected to be :-

(1) The same as the magnetic moment of the cyanide complex

(2) Larger than the magnetic moment of the cyanide complex because there are more unpaired electrons in the fluoride complex

(3) Smaller than the magnetic moment of the cyanide complex because there are more unpaired electrons in the fluoride complex

(4) Larger than the magnetic moment of the cyanide complex because there are fewer unpaired electrons in the fluoride complex

CC0042

47. Which of the following contains one unpaired electron in the 4p orbitals :-

(1) $[Cu(NH_3)_2]^+$ (2) $[Cu(NH_3)_4]^{2+}$

(3) $[Cu(CN)_4]^{3-}$ (4) $[Ni(CN)_4]^{2-}$

CC0043

48. How many unpaired electrons are present in the Brown Ring complex $[Fe(H_2O)_5(NO)]SO_4$

(1) 4 (2) 3 (3) 0 (4) 5

CC0045

49. Which of the following statements about $[Cr(CO)_6]$ is correct ?

(1) It is paramagnetic and inner orbital complex

(2) It is diamagnetic and inner orbital complex

(3) It is diamagnetic and outer orbital complex

(4) It is paramagnetic and outer orbital complex

CC0046

50. Which is true for complex $[Ni(en)_2]^{2+}$?

(1) paramagnetism, dsp^2, square planar,C.N. of Ni = 2

(2) diamagnetism, dsp^2, square planar, C.N. of Ni = 4

(3) diamagnetism, sp^3, tetrahedral, C.N. of Ni = 4

(4) paramagnetism, sp^3, tetrahedral, C.N. of Ni = 4

CC0047

51. Arrange the following in order of decreasing number of unpaired electrons :

I : $[Fe(H_2O)_6]^{2+}$ II : $[Fe(CN)_6]^{3-}$

III : $[Fe(CN)_6]^{4-}$ IV : $[Fe(H_2O)_6]^{3+}$

(1) IV, I, II, III (2) I, II, III, IV

(3) III, II, I, IV (4) II, III, I, IV

CC0048

52. Spin magnetic moment of $[MnBr_4]^{-2}$ is 5.9 BM its geometry would be :-

(1) Square planer (2) Tetrahedral

(3) Square pyramidle (4) Octahedral

CC0240

53. The hybridisation in $[Co(C_2O_4)_3]^{-3}$

(1) sp^3d^2 (2) sp^3d^3

(3) dsp^3 (4) d^2sp^3

CC0241

54. Which complex is diamagnetic ?

(1) $[MnO_4]^-$ (2) $[Co(NH_3)_6]^{+3}$

(3) $[Fe(CN)_6]^{-3}$ (4) $[Cr(H_2O)_6]^{+3}$

CC0242

CFT, COLOUR OF THE COMPLEX AND STABILITY

55. Which one of these ions absorbs energy from visible spectrium :-

(1) $[Cu(NH_3)_4]^+$

(2) $[Cu(NH_3)_4]^{+2}$

(3) $[Zn(H_2O)_6]^{+2}$

(4) $[Co(H_2O)_6]^{+3}$

CC0049

56. In the complex $[Ni(H_2O)_2(NH_3)_4]^{+2}$ the magnetic moment (μ) of Ni is :-

(1) Zero (2) 2.83 BM

(3) 1.73 BM (4) 3.87 BM

CC0050

57. Which of the following system has maximum number of unpaired electrones :-

(1) d^5 (Octahedral, low spin)

(2) d^8 (Tetrahedral)

(3) d^6 (Octahedral, low spin)

(4) d^3 (Octahedral)

CC0051

58. In an octahedral crystal field, the t_{2_g} orbitals are

(1) Raised in energy by 0.4 Δ_0

(2) Lowered in energy by 0.4 Δ_0

(3) Raised in energy by 0.6 Δ_0

(4) Lowered in energy by 0.6 Δ_0

CC0052

59. If $\Delta_0 < P$, the correct electronic configuration for d^4 system will be :-

(1) $t_{2g}^4\ e_g^0$ (2) $t_{2g}^3\ e_g^1$

(3) $t_{2g}^0\ t_g^4$ (4) $t_{2g}^2\ e_g^2$

CC0053

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60. Match List–I (Complex ions) with List–II (Number of Unpaired Electrons) and select the correct answer using the codes given below the lists :

List –I (Complex ions)	**List II (Number of Unpaired Electrons)**
A. $[CrF_6]^{4-}$	i. One
B. $[MnF_6]^{4-}$	ii. Two
C. $[Cr(CN)_6]^{4-}$	iii. Three
D. $[Mn(CN)_6]^{4-}$	iv. Four
	v. Five

Code :	**A**	**B**	**C**	**D**
(1)	iv	i	ii	v
(2)	ii	v	iii	i
(3)	iv	v	ii	i
(4)	ii	i	iii	v

CC0054

61. Consider the following complex formation reactions and comment on their formation constant value

(i) $Fe^{2+}(aq) + 6NH_3 \underset{}{\overset{K_f}{\rightleftharpoons}} \left[Fe(NH_3)_6\right]^{2+}$

(ii) $Fe^{2+}(aq) + 3en \underset{}{\overset{K'_f}{\rightleftharpoons}} [Fe(en)_3]^{2+}$

(1) $K_f > K'_f$ (2) $K_f < K'_f$
(3) $K_f = K'_f$ (4) can not be compared

CC0055

62. Select most stable complex :-
(1) $[Co(H_2O)_6]^{3+}$ (2) $[Co(NH_3)_2(en)_2]^{+3}$
(3) $[Co(en)_3]^{+3}$ (4) $[Co(NH_3)_4(en)]^{+3}$

CC0056

63. Red color of gem stone ruby is due to impurities of:-
(1) Al_2O_3 (2) Cr^{+3}
(3) Al^{+3} ions (4) All of these

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ISOMERISM

64. A square planar complex is cis platin $[Pt(NH_3)_2Cl_2]$ shows :
(1) Geometrical isomerism (2) Optical isomerism
(3) Linkage isomerism (4) Ionisation isomerism

CC0060

65. The two compounds sulphato penta-ammine cobalt (III) bromide and penta-amminebromo cobalt (III) sulphate represent :
(1) Linkage isomerism
(2) Ionisation isomerism
(3) Co-ordination isomerism
(4) Hydrate isomerism

CC0061

66. Which of the following complex can not exhibit geometrical isomerism :-
(1) $[Pt(NH_3)_2Cl\,NO_2]$ (2) $[Pt(gly)_2]$
(3) $[Cu(en)_2]^{+2}$ (4) $[Pt(H_2O)(NH_3)BrCl]$

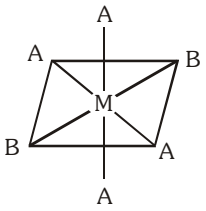
CC0062

67. Which one of the following compounds will exhibit linkage isomerism :-
(1) $[Pt\,(NH_3)_2\,Cl_2]$ (2) $[Co\,(NH_3)_3\,NO_2]Cl_2$
(3) $[Co\,(NH_3)_4\,Cl_2]Cl$ (4) $[Co\,(en)_2Cl_2]Cl$

CC0063

68. The compound $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_4Cl_2]Cl.2H_2O$ respresent
(1) Linkage isomerism (2) Hydrate isomerism
(3) Ligand isomerism (4) Ionisation isomerism

CC0066

69. Compound - 

can be marked as -
(1) Cis form (2) Leavo form
(3) Dextro form (4) Trans form

CC0067

70. A Planar Complex (Mabcd) gives :-
(1) Two Optical isomer
(2) Two geometrical isomer
(3) Three optical isomer
(4) Three geometrical isomers

CC0068

71. No. of geometrical isomers expected for octahedral complex [Mabcdef] is :-
(1) Zero (2) 30 (3) 15 (4) 9

CC0069

72. Which of the following complex produce ppt with $AgNO_3$ and exist in two geometrical isomeric form
(1) $PtCl_2.4NH_3$ (2) $PtCl_2.3NH_3$
(3) $PtCl_4.4NH_3$ (4) $PtCl_4.2NH_3$

CC0070

73. Cis$[Pt(en)_2Br_2]Cl_2.2H_2O$ and trans$[Pt(en)_2Br_2]Cl_2.2H_2O$ are :-
(1) Ionisation isomer (2) Geometrical isomer
(3) Hydrate isomer (4) Optical isomers

CC0071

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74. Which of the following octahedral complex have maximum stereoisomers :-

(1) $[Fe(CN)_6]^{-4}$ (2) $[Co(en)(CN)_4]^{-1}$

(3) $[Co(en)_2(Cl)_2]^{+}$ (4) $[Co(en)_3]^{+3}$

CC0072

75. Which of the following complex does no show geometrical isomerism ?

(1) $[Co\ (NH_3)_4\ Cl_2\ ]^{+}$ (2) $[Co\ (NH_3)_3\ (NO_2)_3]$

(3) $[Cr\ (en)_3\ ]^{3+}$ (4) $[Pt\ (gly)_2]$

CC0073

76. Which of the following complex compounds does not exhibits geometrical isomerism ?

(1) $[PtCl_2(NH_3)_2]$ (2) $[PdCl_2BrI]$

(3) $[Pt(NH_3)\ (py)\ (Cl)\ (Br)]$ (4) $[Pt(NH_3)_3(Br)]$

CC0075

77. The pair of molecules that exhibit geometrical isomerism are

(1) $[\ Cr(NH_3)_4Cl_2]^{+}$and $[Co(NH_3)_3Cl_3]$

(2) $[Cr\ (NH_3)_5Cl]^{2+}$ and $[Co(NH_3)_5Cl]^{2+}$

(3) $[Cr(NH_3)_6]^{3+}$ and $[Co(NH_3)_6]^{3+}$

(4) All of the above show geometric isomerism

CC0076

78. Complex $[Co(NH_3)\ (NO_2)_3]$ show which isomerism :-

(1) Ionisation and linkage

(2) Linkage and geometrical

(3) Geometrical and ligand

(4) Coordination and linkage

CC0243

79. Which of the following represents incorrect match?

(1) $[Co(NH_3)_5(NO_2)](NO_3)_2$: Linkage isomerism

(2) $[Pt\ (NH_3)(H_2O)Cl_2]$: Geometrical isomerism

(3) $K[Cr(H_2O)_2(C_2O_4)_2]$: Geometrical and optical

(4) $[Cr(H_2O)_5Cl]Cl_2H_2O$: Hydrate and linkage isomerism

CC0244

80. $CrCl_3.6H_2O$ show 3 different coloured compounds due to :-

(1) Ionisation isomerism

(2) Coordination isomerism

(3) Optical isomerism

(4) Hydrate isomerism

CC0245

ORGANOMETALLIC COMPOUNDS

81. Which of the following is π –acid ligand

(1) NH_3 (2) CO

(3) glycinato (4) ethylene diamine

CC0077

82. Which of the following statement is/are wrong:-

(a) Al_4C_3 is an organometallic compound

(b) Metal carbonyls are organometallic compounds

(c) TEL is π bonded organometallic compound

(d) Frankland reagent is σ - bonded organometallic compound

The answer is:-

(1) c and d (2) a and c

(3) a and b (4) b & d

CC0078

83. Compounds which contain direct metal carbon bonds are called :

(1) Organic compound (2) Complex compound

(3) Metal carbides (4) OMC compounds.

CC0079

84. Which one of the following is a π bonded OMC ?

(1) Wilkinson's catalyst (2) Tetraethyl lead

(3) Zeise's salt (4) Grignard's reagent

CC0080

85. Which of the following statement is true

(1) $FeCO_3$ and Fe_3C are organometallic compounds.

(2) In ferrocene ligand is cyclopentadienyl.

(3) $Pb\ (C_2H_5)_4$ is π-bonded OMC

(4) In zeise salt central metal is Sp^3 hybridised.

CC0081

86. Which of the following is not an organo metallic compound :-

(1) $(C_2H_5)_2Zn$ (2) $CH_3B(OCH_3)_2$

(3) $B(OCH_3)_3$ (4) $Ni(CO)_4$

CC0084

87. Number of metal -metal bonds in $[Co_2(CO)_8]$ will be :-

(1) 0 (2) 3 (3) 2 (4) 1

CC0246

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APPLICATION OF COORDINATION CHEMISTRY

88. In a ferric salt on adding $K_4[Fe(CN)_6]$ a prussian blue is obtained which is :-
(1) $K_3[Fe(CN)_6]$ (2) $Fe_3[Fe(CN)_6]_4$
(3) $FeSO_4[Fe(CN)_6]$ (4) $Fe_4[Fe(CN)_6]_3$

CC0086

89. Hypo is used in photography because it is :-
(1) A reducing agent
(2) A oxidising agent
(3) A Complexing agent
(4) Photo sensitive Compound

CC0087

90. The solubility of AgBr in hypo solution is due to the formation of :-
(1) Ag_2SO_3 (2) $Ag_2S_2O_3$
(3) $[Ag(S_2O_3)]^-$ (4) $[Ag(S_2O_3)_2]^{3-}$

CC0088

91. Which of the following is related to Nessler's reagent?
(1) $PtCl_4 + KCl \rightarrow$ (2) $AgCl + NH_3 \rightarrow$
(3) $AgBr + Na_2S_2O_3 \rightarrow$ (4) $HgI_2 + KI \rightarrow$

CC0089

92. $K_4[Fe(CN)_6]$ reacts with $FeCl_3$ to form :-
(1) $K_3Fe(CN)_6$ (2) $K_4[Fe(CN)_3Cl_3]$
(3) $K_3[Fe(CN)_5Cl]$ (4) $KFe[Fe(CN)_6]$

CC0090

93. A reagent used for identifying nickel ion is :-
(1) Potassium ferrocyanide
(2) Phenolphthalein
(3) Dimethyl glyoxime
(4) EDTA

CC0091

94. Which one of the following statement is false for nickel-dimethylglyoximate complex ?
(1) The stability of complex is only due to the presence of intra-molecular hydrogen bonding
(2) The complex is stable, only because dimethyl glyoxime ligand is a stronger ligand
(3) The complex is stable as it has five membered chelate rings as well as intra molecular hydrogen bonding
(4) (1) and (2) both

C0092

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EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**1**	**4**	**4**	**2**	**4**	**3**	**4**	**1**	**2**	**2**	**2**	**4**	**2**	**4**	**2**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**3**	**3**	**2**	**1**	**1**	**1**	**1**	**3**	**2**	**2**	**2**	**3**	**4**	**2**	**2**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**3**	**1**	**3**	**4**	**2**	**1**	**1**	**2**	**1**	**1**	**2**	**3**	**3**	**1**	**4**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**2**	**2**	**2**	**2**	**2**	**1**	**2**	**4**	**1**	**1**	**2**	**4**	**2**	**1**	**3**
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	**2**	**3**	**2**	**1**	**2**	**3**	**2**	**2**	**4**	**4**	**3**	**3**	**2**	**3**	**3**
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	**4**	**1**	**2**	**4**	**4**	**2**	**2**	**4**	**3**	**2**	**3**	**4**	**4**	**3**	**4**
Que.	91	92	93	94											
Ans.	**4**	**4**	**3**	**4**											

EXERCISE-III (Analytical Questions) — Master Your Understanding

1. A complex compound of cobalt has molecular formula containing five NH_3 molecules, one nitro group and two chlorine atom for one cobalt atom. One mole of this compound produces three moles of ion in aqueous solution. On reaction with excess of $AgNO_3$ solution two moles of AgCl get precipitated. The Ionic formula of the compound.
(1) $[Co(NH_3)_4 NO_2 Cl] NH_3Cl$
(2) $[Co(NH_3)_5Cl] ClNO_2$
(3) $[Co(NH_3)_5NO_2]Cl_2$
(4) $[Co(NH_3)_5(NO_2)Cl] Cl$

CC0161

2. Which of the following is react with AgCl ?
(1) KCN (2) NH_4OH
(3) $Na_2S_2O_3$ (4) All

CC0162

3. Which one of the following shows maximum paramagnetic character ?
(1) $[Fe(CN)_6]^{3-}$ (2) $[Fe(CN)_6]^{4-}$
(3) $[Cr(H_2O)_6]^{3+}$ (4) $[Cu(H_2O)_6]^{2+}$

CC0163

4. Which statement is incorrect :-
(1) $Ni(CO)_4$ - Tetrahedral, paramagnetic
(2) $[Ni(CN)_4]^{-2}$ - Square planar, diamagnetic
(3) $Ni(CO)_4$ - Tetrahedral, diamagnetic
(4) $[NiCl_4]^{-2}$ - Tetrahedral, paramagnetic

CC0164

5. Considering H_2O as a weak field ligand, the number of unpaired electrons in $[Mn(H_2O)_6]^{2+}$ will be (At. no. of Mn = 25) :-
(1) Four (2) Three (3) Five (4) Two

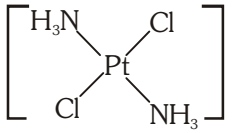
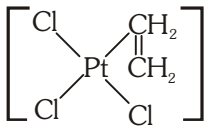
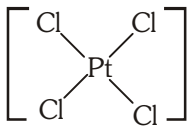
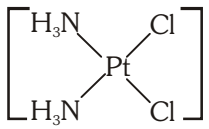
CC0166

6. Which of the following coordination compounds would exhibit optical isomerism ?
(1) tris-(ethylenediamine) cobalt (III) bromide
(2) pentaamminenitrocobalt (III) iodide
(3) diamminedichloroplatinum (II)
(4) trans-dicyanobis (ethylenediamine) chromium (III) chloride

CC0167

7. Among $[Ni(CO)_4]$, $[Ni(CN)_4]^{2-}$, $[NiCl_4]^{2-}$ species, the hybridization states at the Ni atom are, respectively (At. no. of Ni = 28):-
(1) dsp^2, sp^3, sp^3 (2) sp^3, dsp^2, dsp^2
(3) sp^3, dsp^2, sp^3 (4) sp^3, sp^3, dsp^2

CC0168

8. Which of the following is considered to be an anti-cancer species ?
(1) 
(2) 
(3) 
(4) 

CC0169

9. Which of the following does not have a metal-carbon bond ?
(1) $Ni(CO)_4$ (2) $Al(OC_2H_5)_3$
(3) C_2H_5MgBr (4) $K[Pt(C_2H_4)Cl_3]$

CC0170

10. The correct order for the wavelength of absorption in the visible region is :
(1) $[Ni(NO_2)_6]^{4-} < [Ni(NH_3)_6]^{2+} < [Ni(H_2O)_6]^{2+}$
(2) $[Ni(NO_2)_6]^{4-} < [Ni(H_2O)_6]^{2+} < [Ni(NH_3)_6]^{2+}$
(3) $[Ni(H_2O)_6]^{2+} < [Ni(NH_3)_6]^{2+} < [Ni(NO_2)_6]^{4-}$
(4) $[Ni(NH_3)_6]^{2+} < [Ni(H_2O)_6]^{2+} < [Ni(NO_2)_6]^{4-}$

CC0172

11. The IUPAC name of the coordination compound $K_3[Fe(CN)_6]$ is
(1) potassium hexacyanoferrate (II)
(2) potassium hexacyanoferrate (III)
(3) potassium hexacyanoiron (II)
(4) tripotassium hexacyanoiron (II)

CC0173

12. Which one of the following cyano complexes would exhibit the lowest value of spin magnetic moment ?
(1) $[Cr(CN)_6]^{3-}$ (2) $[Mn(CN)_6]^{3-}$
(3) $[Fe(CN)_6]^{3-}$ (4) $[Co(CN)_6]^{3-}$

CC0174

13. A square planar complex is formed by hybridization of which atomic orbitals
(1) s, p_x, p_y, d_{yz} (2) s, p_x, p_y, $d_{x^2-y^2}$
(3) s, p_x, p_y, d_{z^2} (4) s, p_x, p_y, d_{xy}

CC0176

14. The most stable complex among the following is
(1) $K_3[Al(C_2O_4)_3]$ (2) $[Pt(en)_2]Cl_2$
(3) $[Ag(NH_3)_2]Cl$ (4) $K_2[Ni(EDTA)]$

CC0177

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15. Nickel (Z=28) combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[NiX_4]^{2-}$. The number of unpaired electron (s) in the nickel and geometry of this complex ion are, respectively.

(1) one, square planar (2) two, square planar

(3) one, tetrahedral (4) two, tetrahedral

CC0179

16. In $Fe(CO)_5$, the Fe–C bond possesses

(1) ionic character (2) σ – character only

(3) π –character only (4) both σ & π character

CC0180

17. How many EDTA (ethylenediaminetetraacetic acid) molecules are required to make an octahedral complex with a Ca^{2+} ion ?

(1) One (2) Two (3) Six (4) Three

CC0181

18. The "spin-only" magnetic moment [in units of Bohr magneton, (μ_B)] of Ni^{2+} in aqueous solution would be (At. No. Ni= 28)

(1) 0 (2) 1.73 (3) 2.84 (4) 4.90

CC0182

19. The coordination number of Ni^{+2} is = 4

$NiCl_2$ + KCN (Complex) $\rightarrow$ A (Complex)

A + Conc.HCl (Maximun) $\rightarrow$ B(Chloro complex)

The IUPAC name of A and B are

(1) potassium tetracyanonickelate(II), potassium tetrachloronickelate(II),

(2) tetracyanopotassiumnickelate(II), tetrachloro potassiumnickelate (II),

(3) tetracyanonickel (II), tetrachloronickel (II)

(4) potassiumtetracyanonickel (II), potassiumtetrachloronickel (II).

CC0183

20. Which one of the following is tridentate ligand

(1) NO_2^- (2) oxalate ion

(3) glycinate ion (4) dien

CC0184

21. Both Co^{3+} and Pt^{4+} have a coordination number of six. Which of the following pairs of complexes will show approximately the same electrical conductance for their 0 001 M aqueous solutions

(1) $CoCl_3.4NH_3$ and $PtCl_4.4NH_3$

(2) $CoCl_3.3NH_3$ and $PtCl_4.5NH_3$

(3) $CoCl_3.6NH_3$ and $PtCl_4.5NH_3$

(4) $CoCl_3.6NH_3$ and $PtCl_4.3NH_3$

CC0185

22. Which of the following has a square planar geometry

(1) $[PtCl_4]^{2-}$ (2) $[CoCl_4]^{2-}$

(3) $[FeCl_4]^{2-}$ (4) $[NiCl_4]^{2-}$

CC0186

23. Hybridisation of Cr in $[Cr(NH_3)_6]^{+3}$ is

(1) dsp^2 (2) dsp^3

(3) d^2sp^3 (4) sp^3d^2

CC0187

24. Hybridisation and shape of central metal atom of Wilkinson's catalyst is :-

(1) dsp^2, Square planer

(2) sp^3, Tetrahedral

(3) sp^3d, Trigonal bipyramidal

(4) sp^3d^2, Octahedral

CC0188

25. Which of the following is not an organo metalic compound :-

(1) $(C_2H_5)_2Zn$ (2) $CH_3B(OCH_3)_2$

(3) $B(OCH_3)_3$ (4) $Ni(CO)_4$

CC0189

26. Among the ligands NH_3,en, CN^- and CO the correct order of their increasing field strength, is:-

(1) CO < NH_3 < en < CN^-

(2) NH_3 < en < CN^- < CO

(3) CN^- < NH_3 < CO < en

(4) en < CN^- < NH_3 < CO

CC0191

27. Which one of the following complex ions has geometrical isomers ?

(1) $[Co(en)_3]^{3+}$ (2) $[Ni(NH_3)_5Br]^+$

(3) $[Co(NH_3)_2(en)_2]^{3+}$ (4) $[Cr(NH_3)_4(en)]^{3+}$

CC0192

28. Which among the following will be named as dibromidobis (ethylene diamine) chromium (III) bromide ?

(1) $[Cr(en)Br_2]Br$ (2) $[Cr(en)_3]Br_3$

(3) $[Cr(en)_2Br_2]Br$ (4) $[Cr(en)Br_4]^-$

CC0193

29. The octahedral complex of a metal ion M^{3+} with four monodentate ligands L_1, L_2, L_3 and L_4 absorb wavelength in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is :

(1) $L_3 < L_2 < L_4 < L_1$

(2) $L_1 < L_2 < L_4 < L_3$

(3) $L_4 < L_3 < L_2 < L_1$

(4) $L_1 < L_3 < L_2 < L_4$

CC0194

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30. Which of the following statements is(are) correct ?

(1) $[Ag(NH_3)_2]^+$ is linear with sp hybridised Ag^+ ion.

(2) $NiCl_4^{2-}$ has tetrahedral geometry.

(3) $[Cu(NH_3)_4]^{2+}$, $[Pt(NH_3)_4]^{2+}$ and $[Ni(CN)_4]^{2-}$ have dsp^2 hybridisation of the metal ions.

(4) All the above

CC0195

31. Which of the following statements is not true about the complex ion $[Pt(en)_2Cl_2]^{2+}$?

(1) It has two geometrical isomers – cis and trans

(2) Both the cis and trans isomers display optical activity.

(3) Only the cis isomer displays optical acitivity.

(4) Only the cis isomer has non-superimposable mirror image.

CC0196

32. $[Pt(NH_3)(NH_2OH)(NO_2)(py)]^+$ will form how many optical isomers ?

(1) Zero (2) 3 (3) 4 (4) 5

CC0198

33. All the metal ions contains $t_{2g}^6 e_g^0$ configurations. Whicyh of the following complex will be paramagnetic ?

(1) $[FeCl(CN)_4(O_2)]^{4-}$

(2) $K_4[Fe(CN)_6]$

(3) $[Co(NH_3)_6]Cl_3$

(4) $[Fe(CN)_5(O_2)]^{-5}$

CC0200

34. Read the following statements :

(I) Geometrical isomerism is not observed in square planer complexes whereas optical isomerism may be observed.

(II) Square planar complexes of the type Ma_2b_2 exhibit optical isomerism and two geometrical isomers.

(III) Square planar complex of the type Mabcd, shows three isomers - two cis and one trans.

(IV) In fac-$Co(NH_3)_3(NO_2)_3$ the three identical ligands are present on the triangular face of the octahedral.

The option having incorrect statements is :

(1) I, III & IV (2) I & II

(3) II only (4) I, II, III & IV

CC0249

35. Choose the correct statement

(1) The Complex ion does not have plane of symmetry is optically active.

(2) In a coordination entity of the type $[PtCl_2(en)_2]^{+2}$, only cis isomer shows optical activity

(3) Cis-$(CrCl_2(OX)_2]^{-3}$ is optically inactive were trans $[CrCl_2(OX)_2]^{-3}$ is optically active.

(4) Optical isomerism is shown by square planer complexes.

CC0250

36. The complex $[Co(NH_3)_5(NO_2)]Cl_2$

(1) has two forms one is red whereas the other is yellow due to the phenomenon of geometrical isomerism.

(2) has a red form and a yellow form due to phenomenon of ionization isomerism

(3) has a red form and a yellow form due to phenomenon of linkage isomerism

(4) has a red form and a yellow form due to phenomenon of optical isomerism.

CC0251

37. Choose the correct set representing the given statements as true (T) or false (F).

(I) Coordination isomerism arises from the interchange of ligands between cationic and anionic entities present in a complex.

(II) Thiocyanate ligand is an ambidentate ligand.

(III) Ionisation isomerism arises when the counter ion in a complex salt is itself a potential ligand and can displace a ligand which can then becomes the counter ion.

(IV) $[Co(NH_3)_6][Co(CN)_6]$ do not exhibit coordination isomerism

(1) FTTT (2) TTTT

(3) TTTF (4) FTFT

CC0252

38. Select the incorrect statement

(1) The geometry of a complex can be predicted from the knowledge of its magnetic behaviour on the basis of valence bond theory.

(2) $[Co(NH_3)_6]^{+3}$, is a diamagnetic octahedral complex and an inner orbital complex.

(3) $[CoF_6]^{+3}$ is called an outer orbital complex as it uses 4d orbitals in hydridisation.

(4) $[NiCl_4]^{-2}$ & $[Ni(CO)_4]$ both complexes are tetrahedral.

CC0253

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39. Identify the incorrect one
(1) The magnetic behaviour of metal ions having upto three electrons in the d orbitals in their coordination entities is indepent on the nature of ligand.
(2) Complex ions having d^4, d^5, d^6 & d^7 configuration with presence of SFL are low spin complexes.
(3) Magnetic moment of coordination compounds is independent on nature of ligands.
(4) $[Mn(CN)_6]^{-3}$ has magnetic moment of two unpaired electrons whereas $[MnCl_6]^{-3}$ has four unpaired electrons.

CC0254

40. Among the following statements the incorrect one
(1) The crystal field theory is an electrostatic model which considers the metal ligand bond to be ionic.
(2) The five d-orbitals of an isolated gaseous metal atom/ion have same energy.
(3) The degeneracy of d-orbitals is lost if a spherically symmetrical field of negative charges surrounds the metal atom/ion.
(4) The pattern of splitting of d-orditals depends upon the nature of the crystal field.

CC0255

41. Incorrect statement among the following is :
(1) If $\Delta_0 < P$ the fourth electron enters one of e_g orbitals giving the configuration $t_{2g}^3 e_g^1$.
(2) If $\Delta_0 > P$ the fourth electron enters one of t_{2g} orbitals giving the configuration $t_{2g}^4 e_g^0$.
(3) If $\Delta_0 > P$ then e^- configuration of metal ion d^6 will be represented as $t_{2g}^4 e_g^2$.
(4) All of the above

CC0256

42. If the complex ion $[Co(NH_3)_5(H_2O)]^{+3}$ absorbs blue-green radiation; then the colour of the radiation absorbed by $[Co(NH_3)_5(Cl)]^{+2}$ will be :-
(1) Blue (2) Yellow
(3) Violet (4) Green

CC0257

43. $[Cu(H_2O)_4]^{+2}$ appears :
(1) Blue as it absorbs green radiation
(2) Red as it absorbs green radiation
(3) Blue as it absorbs red radiation
(4) Red as it absorbs blue radiation

CC0258

44. Pick the set correctly representing the given statements as true (T) or false (F).
(I) On heating the violet colour of $[Ti(H_2O)_6]Cl_3$ is disappeared.
(II) Red colour of ruby is due to incorporation of octahedral chromium (III) complexes into the alumina lattice.
(III) In emerald, Co^{+3} ions occurpy octahedral sites in the mineral beryl ($Be_3 Al_2 Si_6 O_{18}$)
(IV) On substitution of water in $[Ni(H_2O)_6]^{+2}$ by ethylenediammine (en) then spin magnetic moment is decreased.
(1) TTFF (2) FTTF (3) TTTF (4) TFTT

CC0259

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EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	4	3	1	3	1	3	4	2	1	2	4	2	4	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	1	3	1	4	3	1	3	1	3	2	3	3	4	4
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	
Ans.	2	1	1	2	1	1	3	1	3	3	3	2	3	1	